

- [CPSC 375: High-Performance Computing](#)

[CPSC 375: High-Performance Computing](#)

Spring 2026

- [Syllabus](#)
- [Schedule](#)
- [Upload](#)
- [The Cluster Project](#)

Homework 7

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Categorize each of the following hardware scenarios into one of Flynn's four categories. Briefly explain why based on the instruction and data streams.
 - A. A student writes a C program that uses a for loop to add two arrays on an old single-core Pentium 4 laptop.
 - B. A high-end gaming PC uses an NVIDIA RTX GPU to calculate the lighting for 2 million pixels simultaneously using a single shader function.
 - C. A research cluster at Trinity College has 4 nodes running a heavy MPI simulation.
2. A SIMD unit has 4 ALUs and needs to process a loop of 10 data items:

```
for (i=0; i<10; i++) x[i] += y[i]
```

 - A. How many rounds of execution are required to complete this loop? .
 - B. During the final round, how many ALUs will be actively performing a calculation, and how many will be idle?
3. If you are writing a program for a NUMA system with CPUs, why is it dangerous to let the operating system move your process from CPU 1 to CPU 2 halfway through the calculation?

• Welcome: Sean

- [LogOut](#)

